SIREN: Rapid A/B Testing of Emergency Alerts Using Simulated LLM Residents

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Abstract

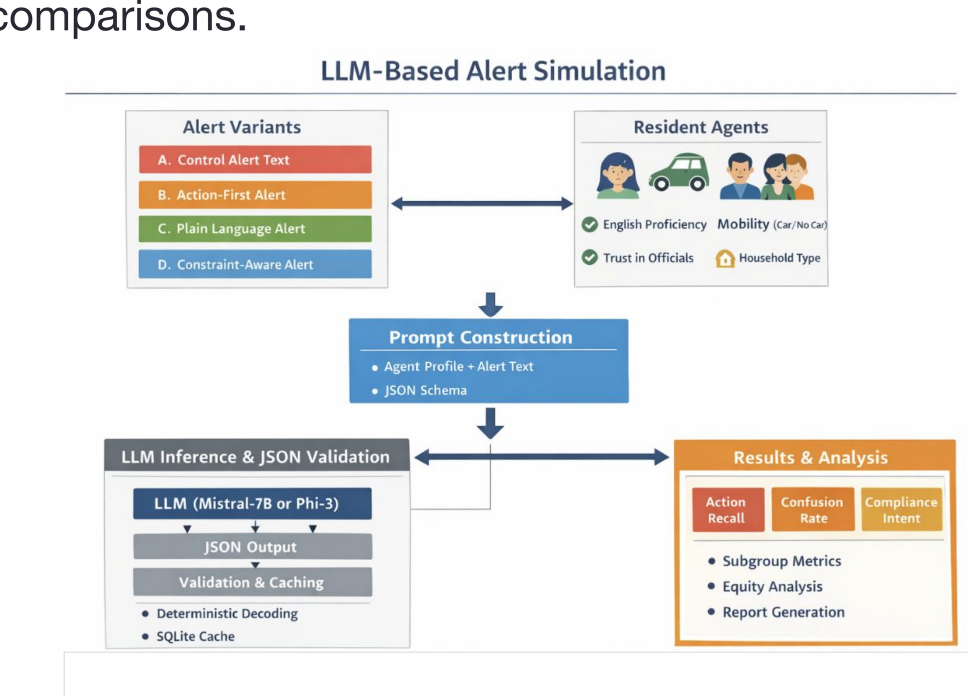
- Emergency alerts must communicate **clear, actionable guidance** under time pressure, yet small differences in wording and structure can affect what people understand and remember.
- We present **SIREN**, a framework that uses **LLM-based "resident agents"** to simulate how different populations respond to alert messages.
- For each alert, we generate **multiple variants** (action-first, plain-language, constraint-aware) and evaluate outcomes such as **action recall, confusion, and intended compliance.**
- Across 17 alerts, we find that **action-first formatting improves recall**, while other variants show **mixed effects**, with differences across **population groups**.
- This approach enables **rapid**, **pre-deployment** testing of emergency alerts, helping identify **effective designs** and **potential failure modes** before real-world use.

Methodology

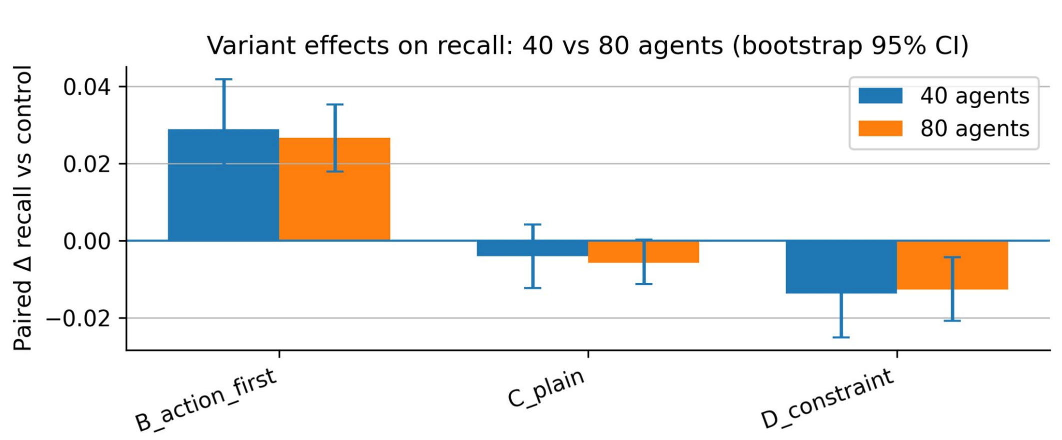
- We evaluated alert effectiveness using **instruction-tuned, open-weight LLMs** in a **GPU-based environment** with **deterministic decoding (temperature = 0)**.
- For each trial, the model outputs **structured JSON responses:**
  - recalled actions
  - confusion
  - intended compliance
- Agents stratified by **English proficiency, mobility, and trust**
- Experiments conducted with **40 and 80 simulated agents**
- **Action recall** computed as fraction of required actions identified
- **Paired comparisons (variant vs. control)** isolate effects
- Results aggregated across subgroups using **bootstrap confidence intervals**

Model Design

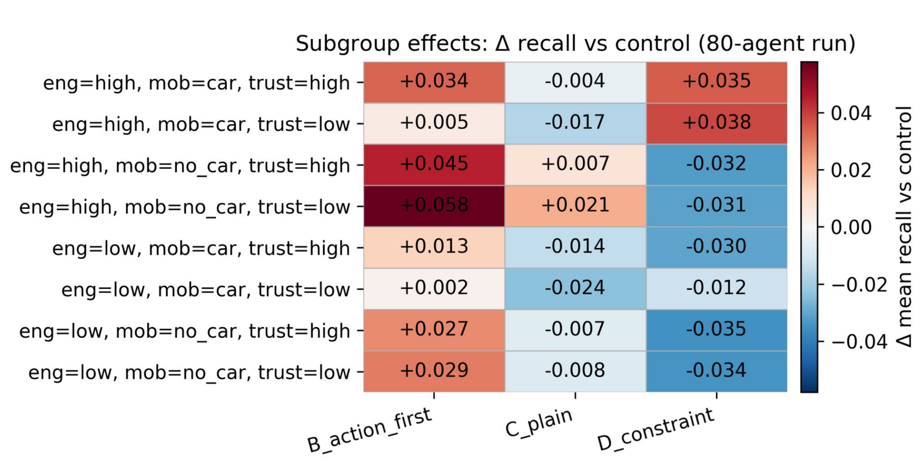
Simulation Design: Alerts are transformed into variants (control, action-first, plain-language, constraint-aware) and evaluated against stratified resident profiles. Each trial returns structured JSON outputs that are scored against human-specified required actions and aggregated for paired A/B comparisons.



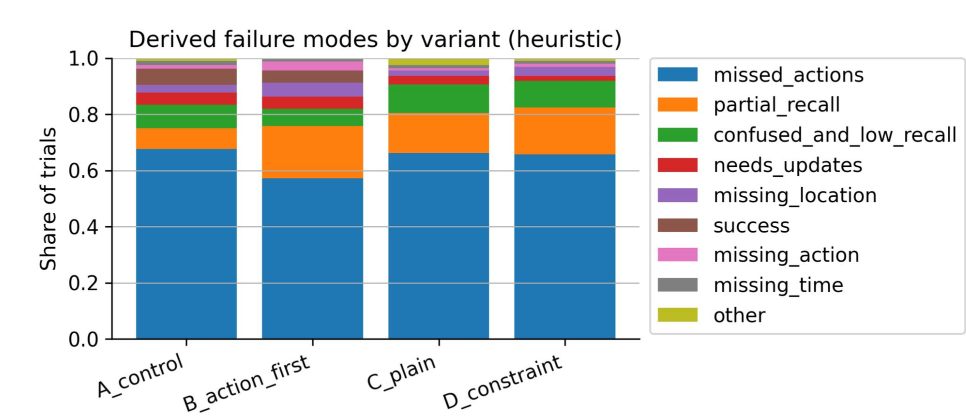
Observations



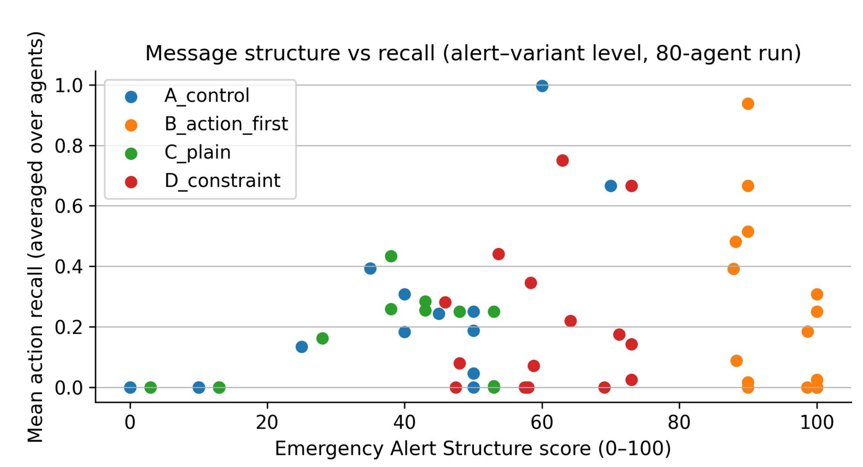
Paired Δ action recall vs. A_control (bootstrap 95% CI), comparing 40-agent and 80-agent runs.



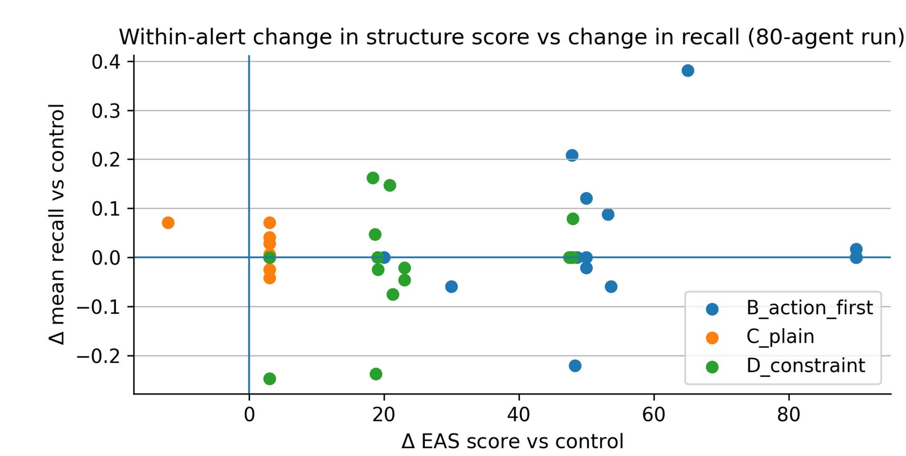
Subgroup heatmap (80-agent run): paired Δrecall vs. control by resident attribute slices.



Derived failure modes by variant (heuristic from trial-level outputs): distribution over missing actions, partial recall, confusion-linked low recall, and other categories.



EAS score vs. mean recall (alert–variant level, 80-agent run).



Within-alert ΔEAS vs. paired Δrecall relative to A_control (80-agent run), colored by variant.

Results

- Across 17 alerts, **action-first formatting** improves recall, showing a consistent positive effect compared to control messages.
- These effects remain **stable across different simulation scales** (40 vs. 80 agents), indicating robustness of the results.
- In contrast, **plain-language and constraint-aware variants show mixed or slightly negative effects**, suggesting that simplification alone does not guarantee better understanding.
- Subgroup analysis reveals **differences across population groups**, highlighting that certain message designs may benefit some users while disadvantaging others.
- The most common failure mode is **missing required actions**, with variations primarily shifting between **partial recall and omission** rather than eliminating errors.
- Higher **message structure and clarity** are associated with improved recall, suggesting that formatting and organization play a key role in effective communication.

Conclusions and Future Works

We conclude that:

- **Action-first message design improves recall,** demonstrating that structure and ordering of information play a critical role in emergency communication.
- **Message effectiveness is not uniform across populations,** with subgroup differences highlighting the importance of equitable alert design.
- **Simplification alone is insufficient,** as plain-language and constraint-aware variants do not consistently improve understanding.
- **Failure modes are driven by missing or partially recalled actions,** rather than complete misunderstanding of messages.
- **Simulation-based evaluation can rapidly surface insights,** enabling faster iteration before real-world deployment.

In the future, we would work on:

- **Validate with human studies** to compare simulated and real-world behavior
- **Improve realism of agents** with richer behavioral and social factors
- **Expand scenarios and alert types** for broader evaluation
- **Incorporate social interactions and information spread**
- **Refine evaluation metrics** for deeper insight into understanding and compliance

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